Residual Seismic Capacity Assessment of Damaged RC Buildings Based on Response Spectrum
Part 2. Assessment of residual seismic capacity ratio based on ideal model and comparison with seismic capacity reduction factor.

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Residual seismic capacity ratio Decline of ductility
Seismic capacity reduction factor Decline of damping
Decline of strength

1. Introduction:

In Part 1, the basic concept of a new method to assess residual seismic capacity by using seismic capacity ratio α has been introduced. Also, the influences of three parameters, which are strength, ductility and equivalent damping on the residual seismic capacity ratio, R, have been investigated. In this part, this method will be applied on ideal behavior models which can combine the three parameters together in assessment of R. Seismic capacity reduction factor which recommended by the code and R in this method will be compared to check the performance of the new method.

2. Definition of R Considering all Three Parameters

According to Part 1, the definitions of seismic capacity ratio α and residual seismic capacity ratio, R, are:

$$\alpha = S_{aL}/S_{as} \quad (1)$$

$$R = \alpha/\alpha_0 \quad (2)$$

where S_{aL} and S_{as} are spectral coordinates of acceleration at the safety limit and seismic response points. α and α_0 are seismic capacity ratios of damaged and undamaged structures, respectively.

For the second kind of design site in Japanese code, where the period $T > 0.864\text{s}$,

$$S_{as} = S_0 G_n F_h = (512/T) \times 2.025 \times 1.5/(1+10h) \quad (3)$$

$$T = 2\pi\sqrt{S_{dL}/S_{aL}} = 2\pi\sqrt{\mu S_{dy}/S_{aL}} \quad (4)$$

where S_0 is the engineering foundation spectrum, G_n is amplification factor for surface site，and h is the equivalent damping of structure. So the residual seismic capacity ratio R becomes

$$R = \sqrt{\mu S_{aL}}\left(1+10h(\mu)\right)\Big/\sqrt{\mu_0 S_{aL0}}\,(1+10h_0) \quad (5)$$

In an SDF system, S_{aL} can be regarded as strength of the structure, and the decline ratios of strength, ductility and equivalent damping were defined as X, Y and Z. So the residual seismic capacity ratio, R, can be

$$R = \sqrt{X}\,\frac{\sqrt{Y} + 3Z\sqrt{Y} - 2.5Z\Big/\sqrt{\mu_0}}{4 - 2.5\Big/\sqrt{\mu_0}} \quad (6)$$

3. Combination Between Strength, Ductility and Damping.

According to the guideline of structural damage assessment and rehabilitation of Japan，the concept of seismic capacity reduction factor, η, can be expressed as area ratio in Fig.1, where η is recommended to assess the residual seismic capacity of structures.

$$\eta = E_d\big/\left(E_r + E_d\right) \quad (7)$$

The capacity curve of the structure which were used in Part 1 to find the seismic capacity ratio, α, will deteriorate as shown in Fig.1. The decline ratio of strength, X, will reduce after the maximum response point exceeds the peak point. The decline ratio of ductility, Y, was produced from residual deformation capacity, u_r. The decline ratio of equivalent damping, Z, was regarded as the decline ratio of energy dissipation.

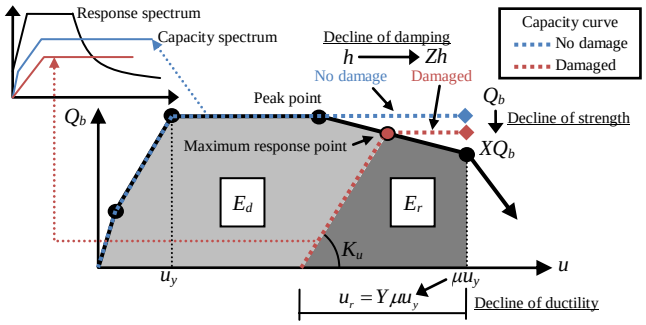

Fig.1 Ideal behavior model of structures

As shown in Fig.2, generally the hysteric loop in the left has good energy dissipation, and the one in the right has poor energy dissipation. Definitions of Y and Z are shown in Fig.2.

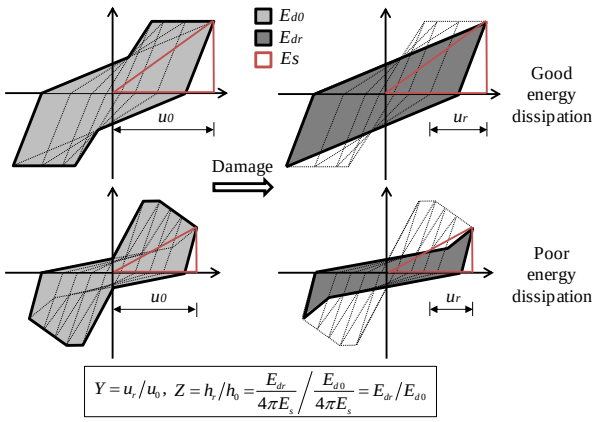

Fig.2 Definition of Y and Z in hysteric loops

As shown in Fig.3, three hysteric loops produced from cyclic loading tests on two RC frames and a coupling beam are shown here as Sample 1, 2 and 3. Sample 1 has most plump hysteric loops and best energy dissipation, so with the decline of residual deformation, Y, the decline of energy dissipation, Z, will be slower. Sample 3 has poorest energy dissipation because of slip and reduction of strength, so the decline of Z will be faster with the decline of Y. Additionally, it was found that Y and Z have an numerical approximate relation as

$$Z = r\ln Y + 1 \quad (8)$$

and poor energy dissipation means a bigger r in Eq.8.

4. Application on Ideal Behavior Model.

Three models with relatively good, moderate and poor ductility were investigated here, as shown in Fig.4. The ultimate ductility

応答スペクトルに基づく被災 RC 造建物の残存耐震性能評価
その２　簡易モデルに基づく耐震残存性能率の評価及び耐震性能低減率の比較

郝　霖霏；姚　倩倩；伊藤　淑紘；鈴木　裕介；前田　匡樹

ratios were 2, 4 and 6 for the three models, and the ultimate strengths was *d* times that of the peak strength, which changed among 0.85, 0.73 and 0.65. For the decline of damping, *r* in Eq.8 has been supposed from 0.40 to 0.80, which represented the energy dissipation varied from relatively good to poor.

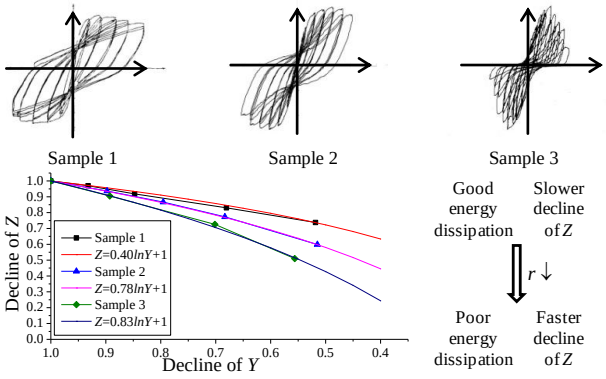

Fig.3 *Y-Z* relationship for different hysteric loops

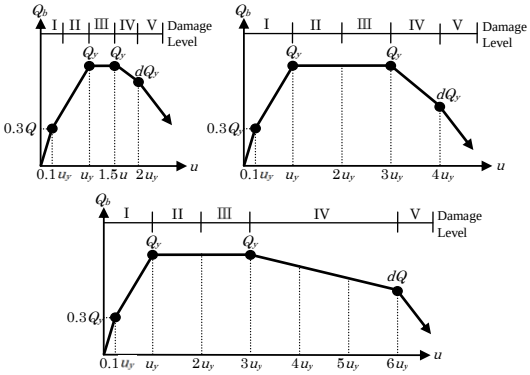

Fig.4 Three behavior models with different ultimate ductility

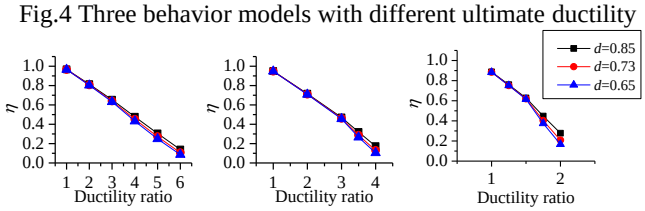

Fig.5 Decline of η in 3 behavior models

As shown in Fig.5, η in the three behavior models were calculated according to Eq.7. It can be found that for different *d* the development of η changed little, and they decline almost uniformly with the increase of ductility ratio, especially when *d* is higher.

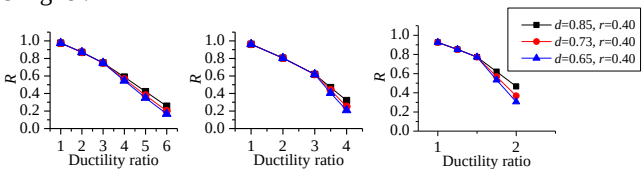

Fig.6 Decline of *R* in three behavior models (constant *r*)

On the other hand, as shown in Fig.6, *R* which was proposed in this paper, was also calculated from these three behavior models. It can be seen that before the reduction of strength, *R* won't change with the variation of *d*. But during the reduction of strength, *R* will be noticeably affected obviously by *d*, and more seriously reducing the strength means *R* will decline faster. It is also shown in Fig.6 that the decline rate of *R* will change before and in the reduction of strength. After the damage level gets into level IV, the decline of *R* will speed up.

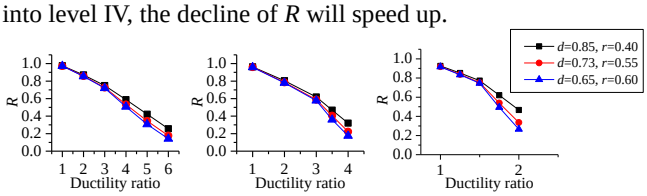

Fig.7 Decline of *R* in three behavior models (varying *r*)

For Fig.6, in each case *r* was assumed to be 0.40, so any change in the decline of damping had not been considered. Generally speaking, faster reduction of strength means poorer energy dissipation, which means a bigger *r* in Eq.8. As shown in Fig.7, *r* was assumed to be bigger for a smaller *d*, and for all behavior models differences between cases became more obvious.

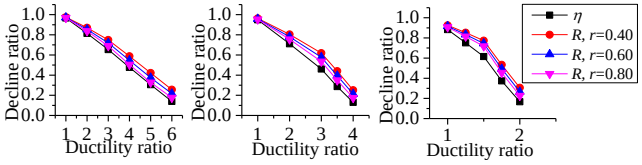

Fig.8 Comparison between η and *R* (varying *r*)

As shown in Fig.8, *d* was supposed to be 0.85 for the first model, and 0.73 for the second model, 0.65 for the third model. For *R*, *r* changed from 0.40 to 0.80, which means energy dissipation of the structure varied from good to poor, and the decline of damping became faster. So η cannot consider the variation of damping, and *R* tends to be bigger than η for either good and poor energy dissipation.

5. Conclusion

(1) The residual seismic capacity ratio, *R*, which is proposed in Part.1 is more sensitive to the reduction of strength and the change between different damage levels, compared with η.

(2) The decline of *R* will be faster when the reduction of strength is more serious and damage level is higher.

(3) *R* will be affected by the different energy dissipation of structures. For structures that have good energy dissipation, *R* will be higher. Also, *R* tends to be higher than η, which cannot consider the influence of energy dissipation and damping of structures.

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